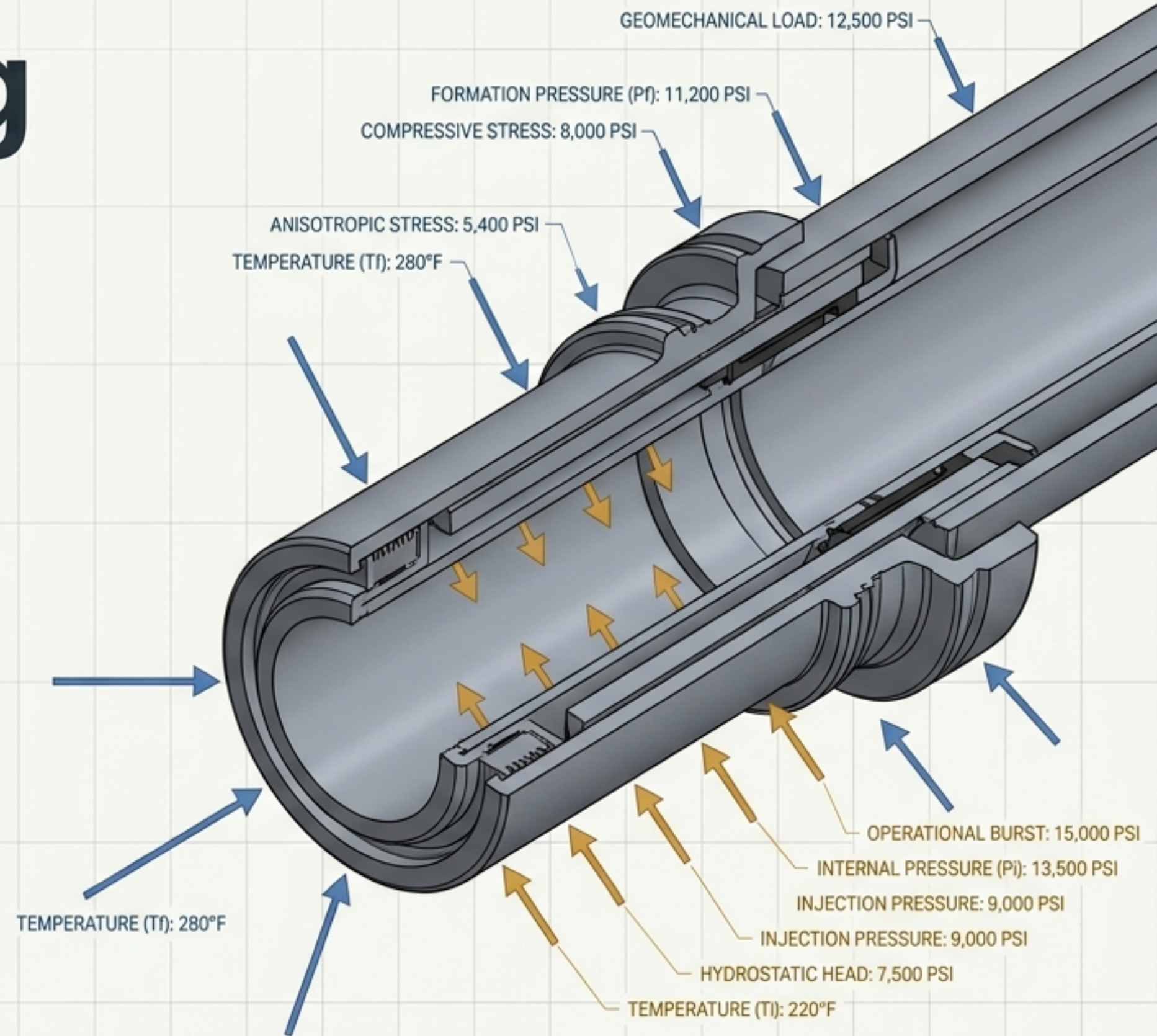


Wellbore Casing Design: Beyond Determinism

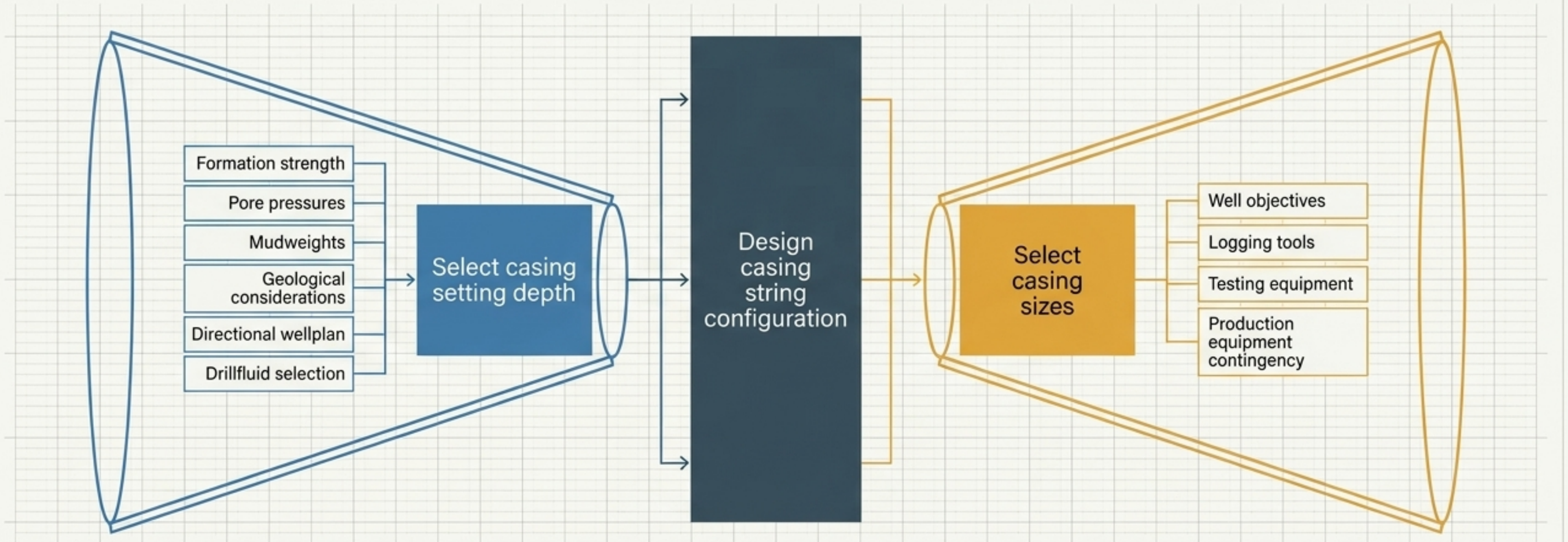
Synthesizing Geomechanical Constraints with Dynamic Hardware Sizing

Prepared by: Eng. Sajjad K. Hano



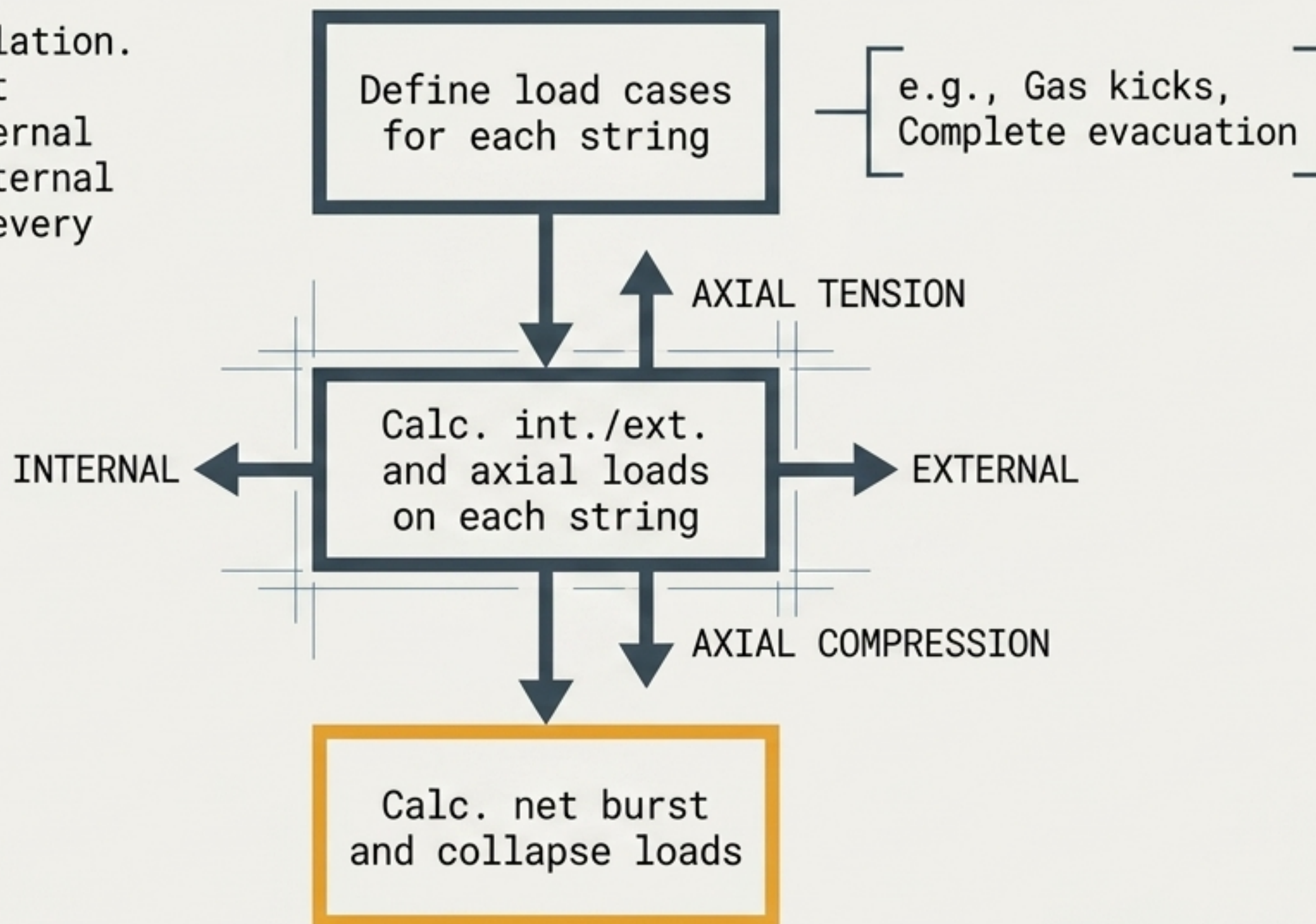
The Dual Funnel of Wellbore Architecture

Casing design begins at the exact intersection of subsurface reality and surface objectives.



Defining the Load Architecture

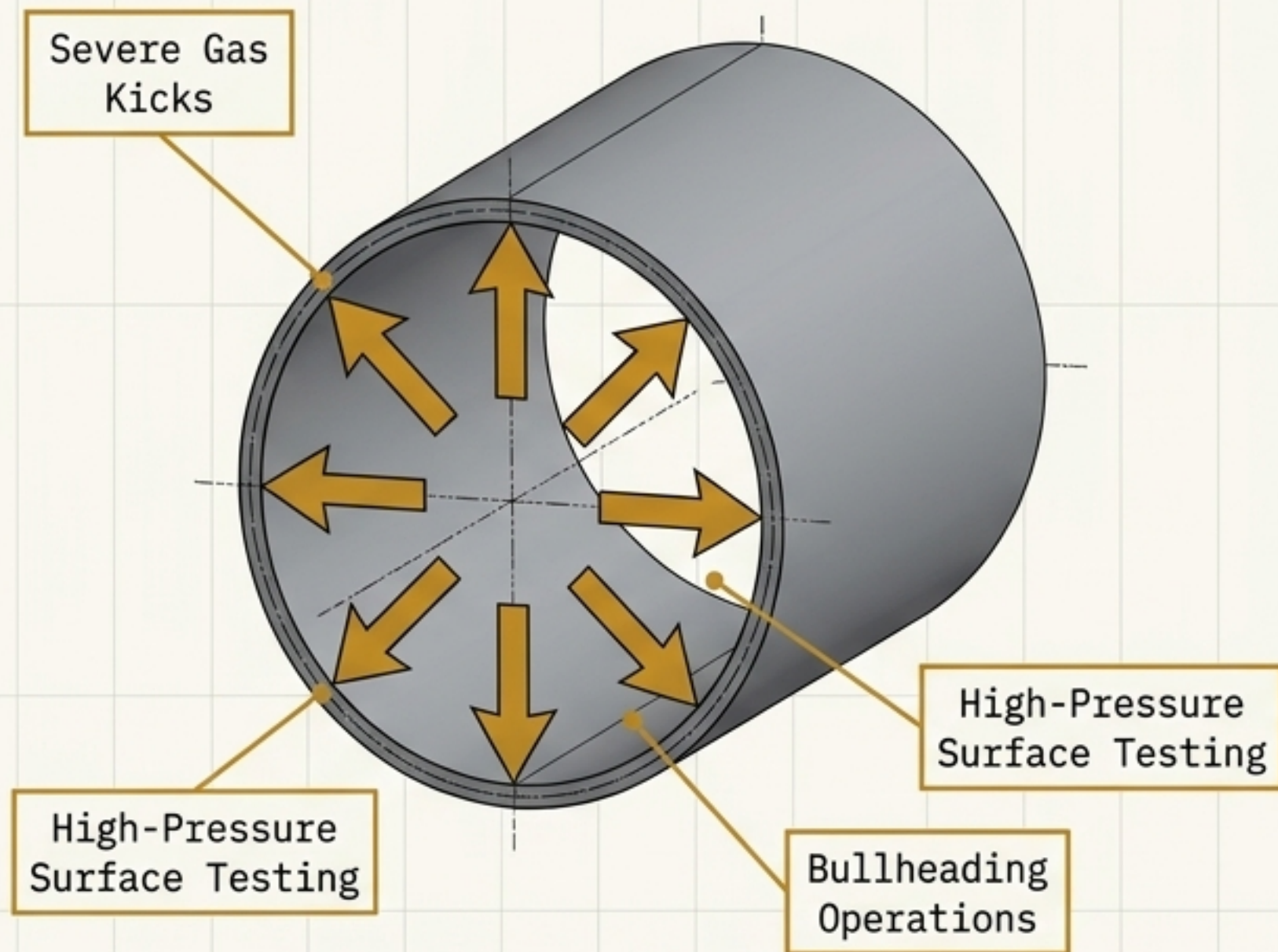
Forces rarely act in isolation. We must calculate the net differential between internal wellbore dynamics and external formation pressures for every string.



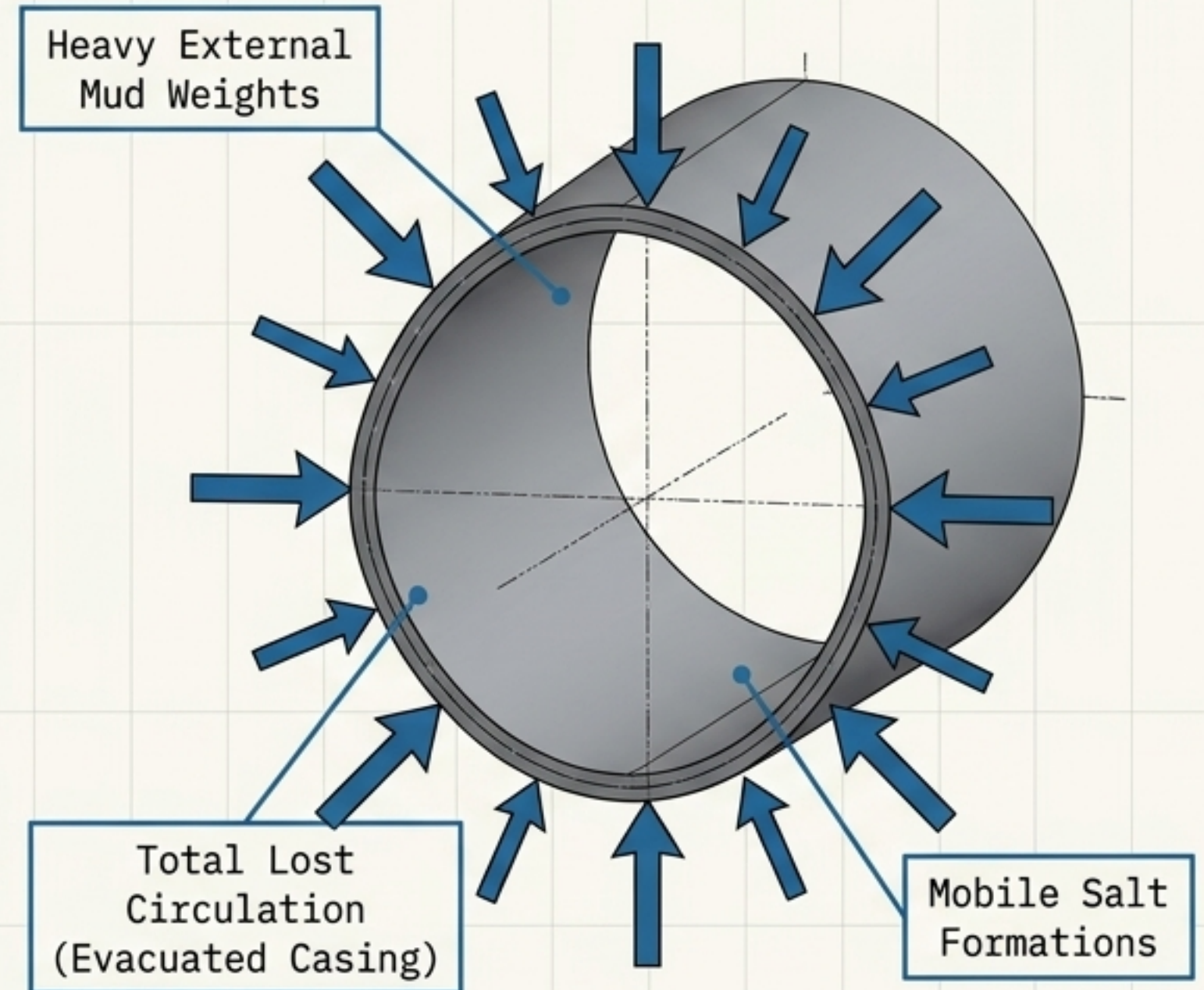
Net Load Profiling: Burst vs. Collapse

The extremes of operational reality dictate the primary failure modes.

Net Burst

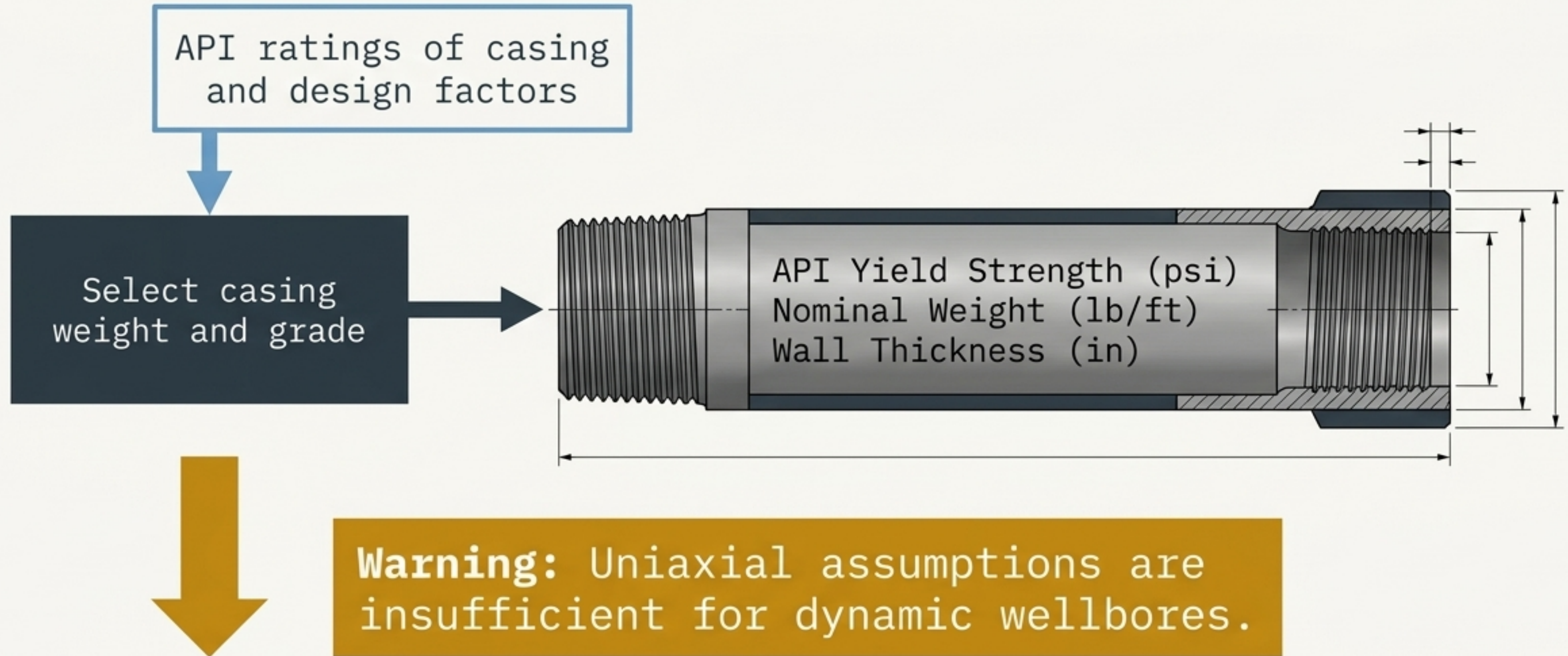


Net Collapse



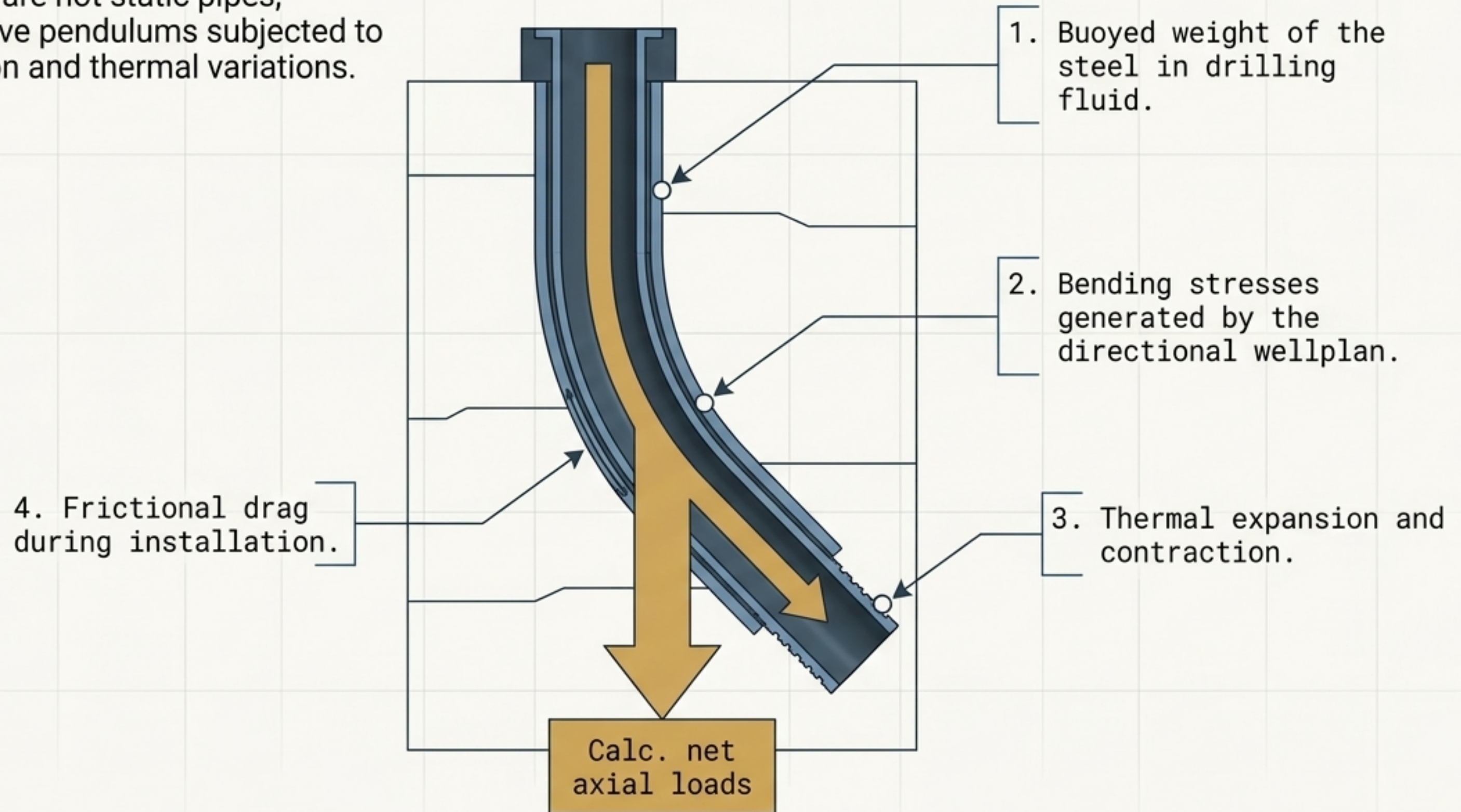
Establishing the Hardware Baseline

Initial selection pairs theoretical net loads against standardized metallurgical capacities.



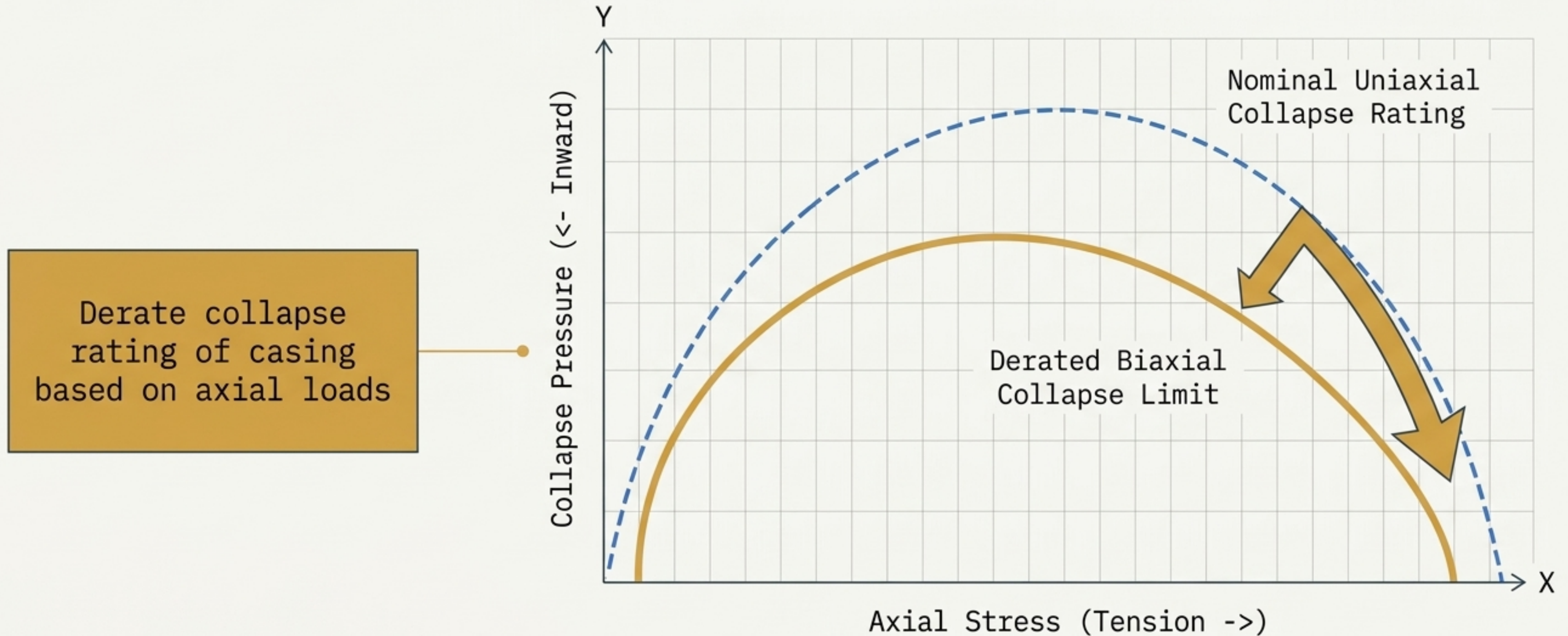
The Reality of Net Axial Loading

Casing strings are not static pipes;
they are massive pendulums subjected to
extreme tension and thermal variations.



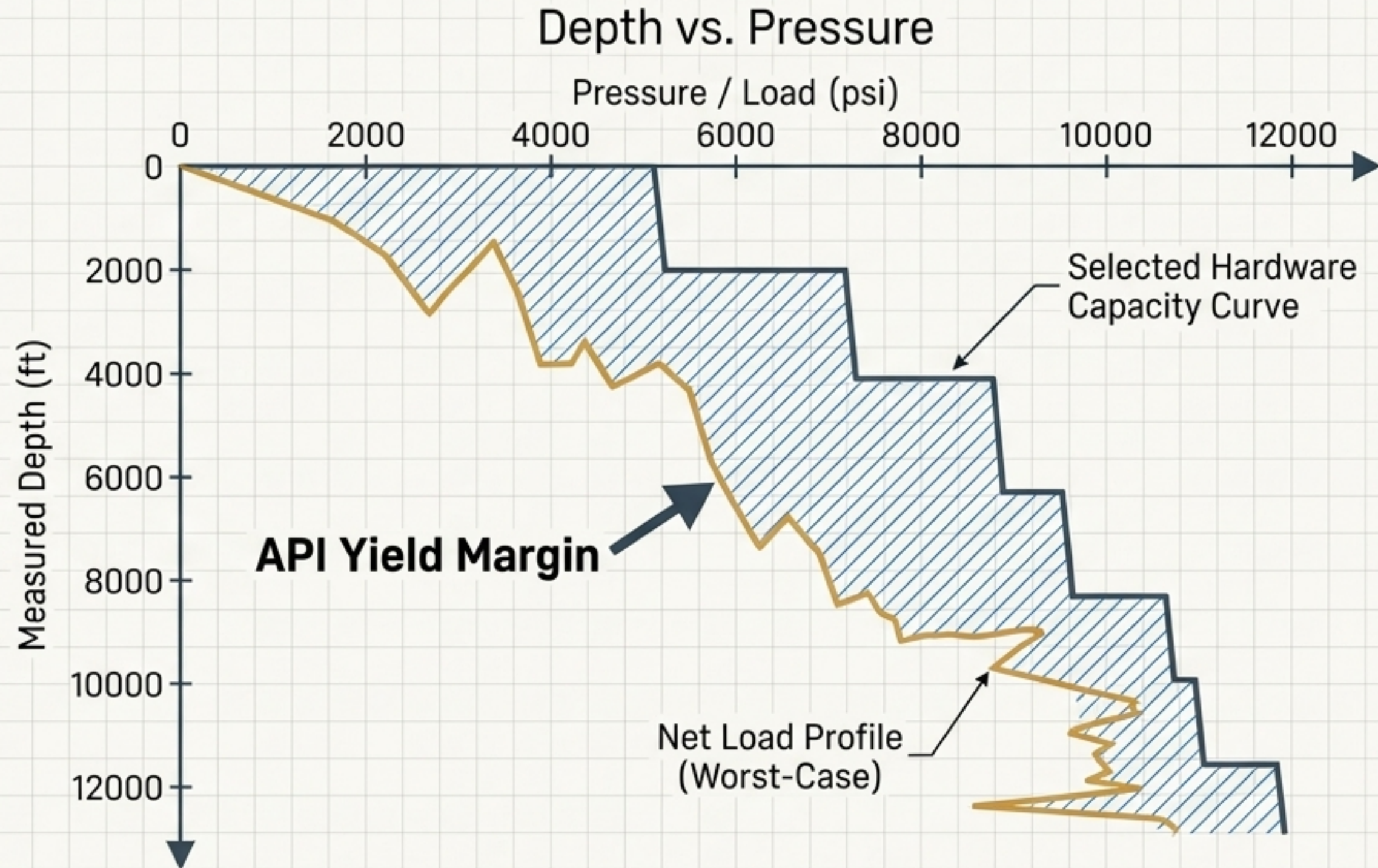
Biaxial Derating: The Von Mises Yield Envelope

Axial tension fundamentally alters metallurgical structure, requiring aggressive derating of collapse resistance.



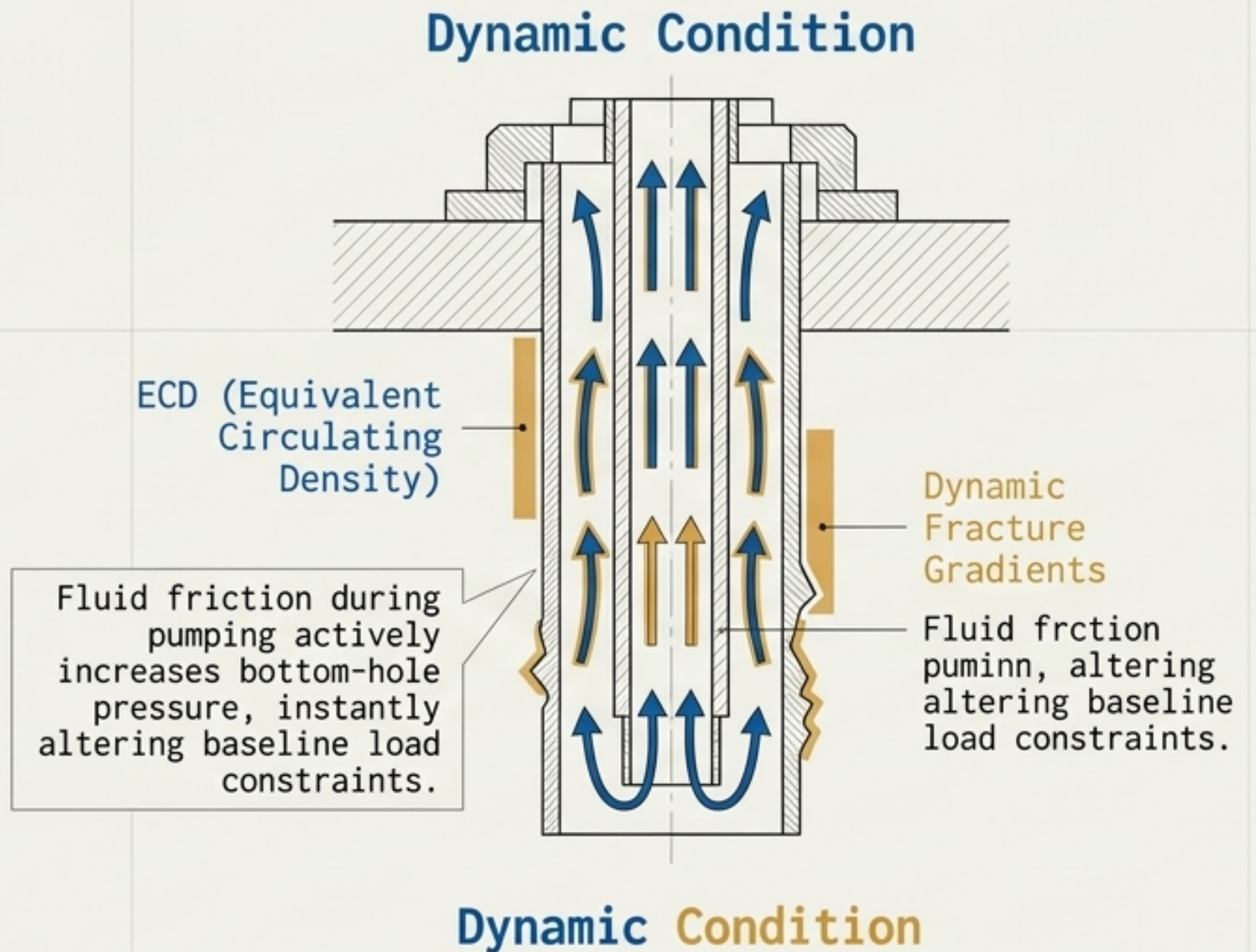
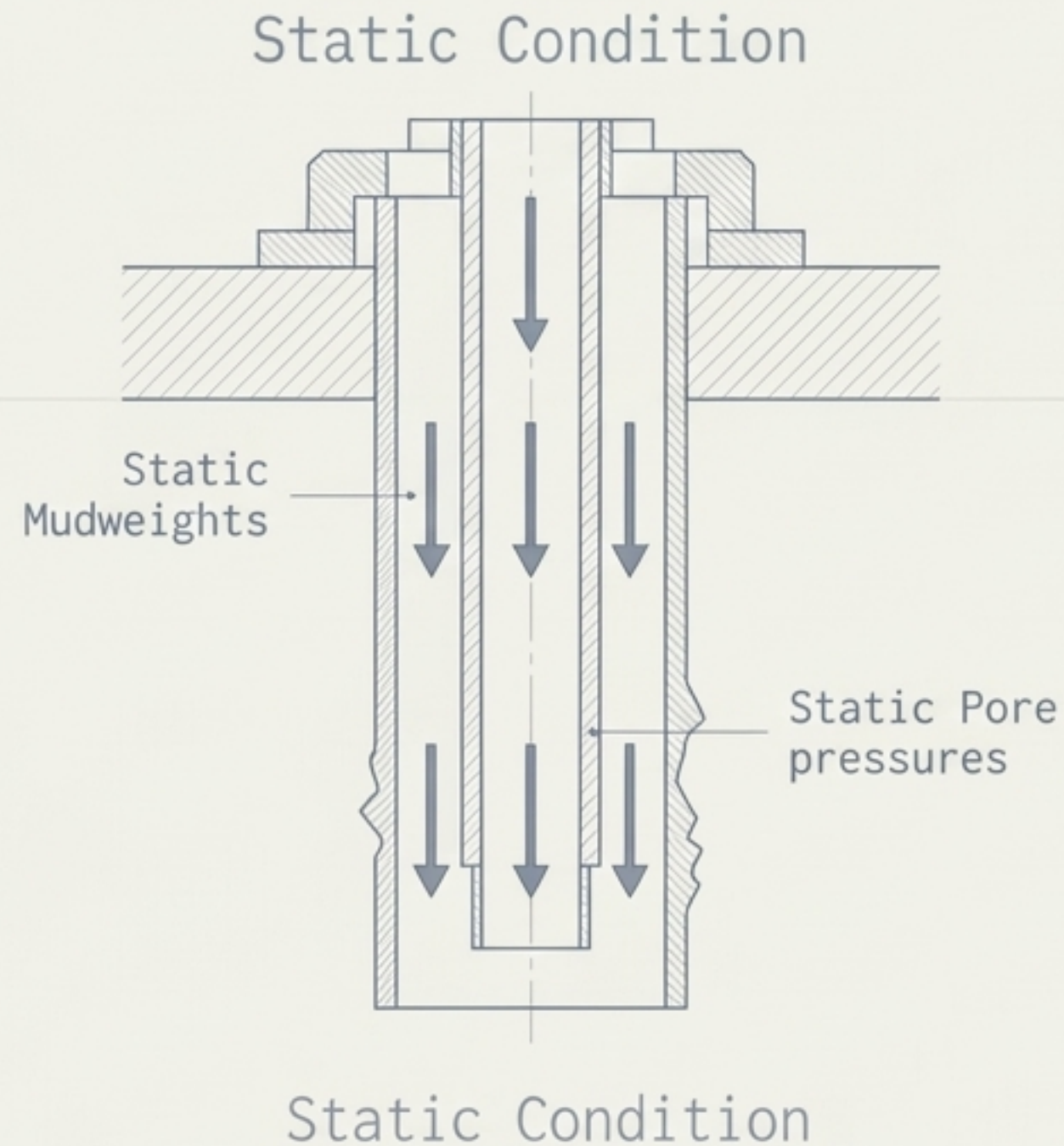
Mapping the True API Yield Margin

Safety is defined by the space between dynamic geological loads and derated metallurgical capacity.



Breaking the Static Assumption

A deterministic model ends at selection. A dynamic model constantly recalibrates against real-time subsurface physics.



Evaluating the Engineering Paradigms

Moving from isolated calculations to a probabilistic, systems-level architecture.

		Legacy Deterministic Model	Integrated Dynamic Workflow
Context Label	Pressure Inputs	Static Mudweights	ECD & Real-time Transient Pressures
	Load Cases	Isolated & Absolute	Probabilistic & Coupled
	Collapse Criteria	Uniaxial API Baseline	Von Mises Biaxial Derating
	Flexibility	Rigid & Sequential	Adaptive Operational Feedback

Closing the Loop: The Dynamic Casing Workflow

Engineering does not stop at selection; it evolves through continuous operational validation.

Live ECD and drilling data continuously validate the API yield margins, ensuring the digital twin perfectly matches subsurface reality.

